In [2]:
```
import numpy as np
import pandas as pd
df=pd.read_csv(r"C:\Users\Public\Downloads\feature scaling dataset.csv")
df
```

Out[2]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Country</th> <th>Age</th> <th>Salary</th> <th>Purchased</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> </tbody> <tbody> </tbody>

	France	44.0	72000	No
	Spain	27.0	48000	Yes
	Germany	30.0	54000	No
	Spain	38.0	61000	No
	Germany	40.0	Nan	Yes
	France	35.0	58000	Yes
	Spain	NaN	52000	No
	France	48.0	79000	Yes
	NaN	50.0	83000	No
	France	37.0	67000	Yes

In [3]:
```
df.head()
```

Out[3]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Country</th> <th>Age</th> <th>Salary</th> <th>Purchased</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

	France	44.0	72000	No
	Spain	27.0	48000	Yes
	Germany	30.0	54000	No
	Spain	38.0	61000	No
	Germany	40.0	Nan	Yes

In [ ]:
```
```

In [8]:
```
label=df.iloc[:,-1].values
from sklearn.impute import SimpleImputer
age=SimpleImputer(strategy="mean",missing_values=np.nan)
Salary=SimpleImputer(strategy="mean",missing_values=np.nan)
age.fit(features[:,[1]])
```

```
SimpleImputer()

Salary.fit(features[:,[2]])

SimpleImputer()

SimpleImputer()

SimpleImputer()

features[:,[1]]=age.transform(features[:,[1]])
features[:,[2]]=Salary.transform(features[:,[2]])
features
```

Out[8]:
```
array([['France', 44.0, 72000.0],
       ['Spain', 27.0, 48000.0],
       ['Germany', 30.0, 54000.0],
       ['Spain', 38.0, 61000.0],
       ['Germany', 40.0, 63777.77777777778],
       ['France', 35.0, 58000.0],
       ['Spain', 38.77777777777778, 52000.0],
       ['France', 48.0, 79000.0],
       [nan, 50.0, 83000.0],
       ['France', 37.0, 67000.0]], dtype=object)
```

In [9]:
```
from sklearn.preprocessing import OneHotEncoder
oh = OneHotEncoder(sparse_output=False)
Country=oh.fit_transform(features[:,[0]])
Country
```

Out[9]:
```
array([[1., 0., 0., 0.],
       [0., 0., 1., 0.],
       [0., 1., 0., 0.],
       [0., 0., 1., 0.],
       [0., 1., 0., 0.],
       [1., 0., 0., 0.],
       [0., 0., 1., 0.],
       [1., 0., 0., 0.],
       [0., 0., 0., 1.],
       [1., 0., 0., 0.]])
```

In [10]:
```
final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)
final_set
```

Out[10]:
```
array([[1.0, 0.0, 0.0, 0.0, 44.0, 72000.0],
       [0.0, 0.0, 1.0, 0.0, 27.0, 48000.0],
       [0.0, 1.0, 0.0, 0.0, 30.0, 54000.0],
       [0.0, 0.0, 1.0, 0.0, 38.0, 61000.0],
       [0.0, 1.0, 0.0, 0.0, 40.0, 63777.77777777778],
       [1.0, 0.0, 0.0, 0.0, 35.0, 58000.0],
       [0.0, 0.0, 1.0, 0.0, 38.77777777777778, 52000.0],
       [1.0, 0.0, 0.0, 0.0, 48.0, 79000.0],
       [0.0, 0.0, 0.0, 1.0, 50.0, 83000.0],
       [1.0, 0.0, 0.0, 0.0, 37.0, 67000.0]], dtype=object)
```

In [11]:
```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)
feat_standard_scaler
```

Out[11]:
```
array([[ 1.22474487e+00, -5.00000000e-01, -6.54653671e-01,
        -3.33333333e-01,  7.58874362e-01,  7.49473254e-01],
       [-8.16496581e-01, -5.00000000e-01,  1.52752523e+00,
        -3.33333333e-01, -1.71150388e+00, -1.43817841e+00],
       [-8.16496581e-01,  2.00000000e+00, -6.54653671e-01,
        -3.33333333e-01, -1.27555478e+00, -8.91265492e-01],
       [-8.16496581e-01, -5.00000000e-01,  1.52752523e+00,
        -3.33333333e-01, -1.13023841e-01, -2.53200424e-01],
       [-8.16496581e-01,  2.00000000e+00, -6.54653671e-01,
        -3.33333333e-01,  1.77608893e-01,  6.63219199e-16],
       [ 1.22474487e+00, -5.00000000e-01, -6.54653671e-01,
        -3.33333333e-01, -5.48972942e-01, -5.26656882e-01],
       [-8.16496581e-01, -5.00000000e-01,  1.52752523e+00,
        -3.33333333e-01,  0.00000000e+00, -1.07356980e+00],
       [ 1.22474487e+00, -5.00000000e-01, -6.54653671e-01,
        -3.33333333e-01,  1.34013983e+00,  1.38753832e+00],
       [-8.16496581e-01, -5.00000000e-01, -6.54653671e-01,
         3.00000000e+00,  1.63077256e+00,  1.75214693e+00],
       [ 1.22474487e+00, -5.00000000e-01, -6.54653671e-01,
        -3.33333333e-01, -2.58340208e-01,  2.93712492e-01]])
```

In [12]:
```
from sklearn.preprocessing import MinMaxScaler
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)
feat_minmax_scaler
```

Out[12]:
```
array([[1.        , 0.        , 0.        , 0.        , 0.73913043,
        0.68571429],
       [0.        , 0.        , 1.        , 0.        , 0.        ,
        0.        ],
       [0.        , 1.        , 0.        , 0.        , 0.13043478,
        0.17142857],
       [0.        , 0.        , 1.        , 0.        , 0.47826087,
        0.37142857],
       [0.        , 1.        , 0.        , 0.        , 0.56521739,
        0.45079365],
       [1.        , 0.        , 0.        , 0.        , 0.34782609,
        0.28571429],
       [0.        , 0.        , 1.        , 0.        , 0.51207729,
        0.11428571],
       [1.        , 0.        , 0.        , 0.        , 0.91304348,
        0.88571429],
       [0.        , 0.        , 0.        , 1.        , 1.        ,
        1.        ],
       [1.        , 0.        , 0.        , 0.        , 0.43478261,
        0.54285714]])
```

In [ ]: